

Ya-Wen (Yama) Chang

yama.chang.gr@dartmouth.edu | yamaychang.me | LinkedIn | GitHub

I am a PhD researcher building AI systems for real-time mental health support. I combine technical expertise in LLM-based systems, multimodal sensing, and adaptive interventions with deep domain knowledge in clinical psychology to design solutions grounded in real clinical needs.

Technical Skills

Languages	Python (primary), R, JavaScript, SQL
ML & AI	TensorFlow/Keras, Scikit-learn, LLM prompt engineering, vLLM, multi-agent systems, time-series modeling
Health & Sensing	EEG/IMU signal processing, multimodal sensing pipelines (GPS, accelerometer, screen), BLE wearable interfacing, spectral analysis
Tools & Infra	Git/GitHub, multi-GPU serving, async processing, Jupyter, Chart.js, Leaflet.js

Technical Projects

Dartmouth College • Center for Technology and Behavioral Health • github.com/yamachang

Neuro-Symbolic Sensor-to-Intervention Pipeline — *Generative AI System for Wellbeing*

Designed and built a hybrid neuro-symbolic architecture to solve a key limitation in LLM-driven interventions: LLM tokenizers cannot natively process numerical sensor time-series (the modality gap), leading to hallucinated clinical reasoning. Developed a semantic tokenization pipeline that translates raw multimodal sensor streams into human-readable behavioral tokens, enabling LLM abductive reasoning over 24-hour context windows. Scaled to **287K+ decision windows across 49 users** on a 4-GPU vLLM cluster.

LLM Decision Engine Evaluation — *Generative AI System for Wellbeing*

Designed an evaluation framework for assessing LLM-based JITA decision reliability across 287K aligned decision-window pairs and 49 users, with multi-level agreement metrics and permutation baselines. Demonstrated **topic agreement 71%**, personalization cosine similarity **0.905**, and agreement **31 σ above chance**. Built a 6-dimension automated evaluation suite.

NeuroZen: Real-Time Closed-Loop EEG Meditation System — *IRB-Approved*

Built an end-to-end closed-loop EEG meditation system in 10 weeks, from data collection to deployment. Designed a real-time pipeline that streams EEG/IMU data via BLE, processes biosignals, and classifies three meditation states (rest/light/deep) using LSTM models with **67% accuracy**—a novel task with no standardized ground truth. System delivers adaptive audio feedback with **latency under 520 ms** and includes a live visualization dashboard. IRB-approved 6-session crossover protocol.

Experience

Dartmouth College

PhD Researcher, AIM HIGH Lab (AI and Mental Health), advised by Nicholas Jacobson

Hanover, NH

Sep. 2024 – Present

Lead researcher on Evergreen, an AI-driven mental wellbeing platform for Dartmouth's student body, with a 600-user evaluation study underway. Collaborate cross-functionally with engineers, designers, and product teams to translate research findings into production features. Define intervention logic and safety requirements that directly inform system architecture and deployment decisions.

Northwestern University

Data Scientist, Lab for Scalable Mental Health

Chicago, IL

Sep. 2022 – Aug. 2024

Built data pipelines and applied machine learning to digital mental health intervention research. Developed a county-level structural stigma index from 600,000+ public records across multiple data sources. Led data analysis in R and Python, including time-series modeling and ML-based prediction. Served as statistical consultant across the lab.

University of Pittsburgh School of Medicine

Senior Research/Data Analyst, Department of Psychiatry

Pittsburgh, PA

Sep. 2020 – Aug. 2022

Led independent research on late-life suicide and depression, building statistical models in R for classification and prediction. Generated data-driven reports to inform recruitment for a case-control study, reducing the gender gap to 8%.

Columbia University Irving Medical Center

Research Intern, Program for the Study of LGBT Health

New York, NY

Sep. 2018 – Mar. 2020

Led an independent research project on LGBTQ+ mental health from conceptualization through quantitative analysis to first-authored manuscript.

Education

Ph.D., Quantitative Biomedical Science	— Dartmouth College, Geisel School of Medicine	2024 –
M.A., Clinical Psychology (Quantitative Methods track)	— Columbia University, GPA: 4.0/4.0	2018 – 2020
B.A., Economics	— National Taiwan University	2008 – 2012

Publications

[Google Scholar Profile](#) (8 peer-reviewed, 2 in preparation) * indicates equal contribution

IN PREPARATION

[P1] **Chang, Y.**, Han, H.J., Griffin, T.Z., Marsch, L., Heinz, M.V., Campbell, A.T., & Jacobson, N.C. (In Preparation). Reliability of an agentic LLM JITAL engine for evidence-based interventions from passive sensing.

[P2] **Chang, Y.**, Heinz, M.V., Griffin, T.Z., Nemesure, M.D., Lekkas, D., Pillai, A., Collins, A.C., Nepal, S., Mackin, D.M., Price, G.D., Campbell, A.T., & Jacobson, N.C. (In Preparation). The digital balance: Using passive sensing of online and in-person social behavior to model daily depressive symptoms.

SELECTED PEER-REVIEWED PUBLICATIONS

[J1] **Chang, Y.**, Sotomayor, I., Szkody, E., Fox, K., & Schleider, J. (2025). Effectiveness of an online single-session minority stress intervention: No evidence for variation by structural stigma, internalized stigma, or social support. *SSM–Mental Health*, 100409.

[J2] Brown, V., Gujral, S., **Chang, Y.**, Galfalvy, H., Szanto, K., & Dombrovski, A.Y. (2025). Brain volume reductions and relationship with depression and cognitive functioning in suicide in older adults: A cross-sectional and longitudinal study using Bayesian multilevel modeling. *Biological Psychiatry: Global Open Science*.

[J3] **Chang, Y.**, Buerke, M., Galfalvy, H., & Szanto, K. (2023). Childhood trauma is associated with early-onset but not late-onset suicidal behavior in late-life depression. *International Psychogeriatrics*, 1–14. PMID: 37642013

[J4] Szkody, E., **Chang, Y.**, & Schleider, J.L. (2023). Serving the underserved? Uptake, effectiveness, and acceptability of digital SSIs for rural American adolescents. *Journal of Clinical Child and Adolescent Psychology*.

Selected Conference Talk

[S1] **Chang, Y.**, Han, H.J., Ross, M.K., Seo, H., & Kaveladze, B. (2025, December). Building an intelligent mental health app: Integrating passive sensing, prediction, and conversational AI interventions. [Symposium presentation]. 9th Annual Technology in Psychiatry Summit (TiPS), Virtual.